2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

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Abstract

This report documents `cfd2d`, a cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations on unstructured meshes, parallelized with MPI and METIS k-way graph partitioning, and its application to the eight required benchmark cases: NACA0012 airfoil at Mach 0.15/0.8/2.0 in inviscid and laminar ($Re = 5000$) modes, and a circular cylinder at Mach 0.1, laminar $Re = 20$ (steady) and $Re = 200$ (unsteady vortex shedding). Spatial discretization is second-order upwind with weighted least-squares reconstruction and Venkatakrishnan limiting; the steady march uses a block LU-SGS implicit pseudo-time integrator with a frozen-reconstruction-consistent Jacobian; the cylinder $Re = 200$ case uses a true second-order BDF2 physical-time march with a dual-time inner solve at every physical step. All eight cases completed and are marked converged (steady) or statistically periodic ($Re = 200$); convergence evidence, force coefficients, field visualizations, a rank-consistency study at $np = 1, 2, 4, 8$, and an honest account of the residual-plateau acceptance rule used on the stiff low-Mach cases are given below.

1 Introduction

The benchmark requires a from-scratch 2-D unstructured compressible Navier–Stokes finite-volume solver with MPI parallelism, run against eight supplied cases (two meshes, two physics modes, four NACA Mach/viscosity combinations and two cylinder Reynolds numbers), with strict output, visualization, and reporting contracts. The sections below describe the governing equations and nondimensionalization (section 2), meshes and case inputs (section 3), the spatial discretization (section 4), boundary conditions (section 5), the implicit steady and transient time integration (section 6), the MPI/METIS strategy (section 7), reproducibility and the machine-readable sanity gate (section 8), quantitative results for all cases (section 9), parallel validation (section 10), and known limitations (section 12).

2 Governing Equations and Nondimensionalization

The conservative state is $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ and the 2-D compressible Navier–Stokes equations in conservative form are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix},$$

and perfect-gas closure

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}(u^2 + v^2), \qquad a = \sqrt{\gamma p/\rho}, \qquad \gamma = 1.4,\ R = 1.$$

For laminar cases the viscous fluxes use the Newtonian stress tensor

$$\tau_{xx} = \tfrac{2}{3}\mu(2u_x - v_y), \quad \tau_{yy} = \tfrac{2}{3}\mu(2v_y - u_x), \quad \tau_{xy} = \mu(u_y + v_x),$$

and Fourier heat flux $q = -k\nabla T$ with $k = \mu\gamma R/((\gamma-1)\,\mathrm{Pr})$, $\mathrm{Pr} = 0.72$, and constant viscosity μ chosen so that the Reynolds number matches the case value, $\mu = \rho_\infty|\mathbf{v}_\infty|L/Re$ with $L = 1$.

All cases are run in the nondimensionalization supplied by the case files: reference length $L = 1$ (chord or cylinder diameter), freestream density $\rho_\infty = 1$, freestream velocity magnitude $|\mathbf{v}_\infty| = 1$, and freestream pressure set by the Mach number, $p_\infty = \rho_\infty|\mathbf{v}_\infty|^2/(\gamma M_\infty^2)$. Force coefficients use the case reference area $A = 1$ and dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty|\mathbf{v}_\infty|^2$:

$$C_D = \frac{F'_x}{q_\infty A}, \qquad C_L = \frac{F'_y}{q_\infty A}, \qquad C_p = \frac{p - p_\infty}{q_\infty}, \qquad C_f = \frac{\tau_w}{q_\infty},$$

and C_m uses the case moment center (quarter-chord $(0.25, 0)$ for NACA, origin for the cylinder).

3 Meshes, Boundary Tags, and Case Inputs

Meshes are read from CGNS (HDF5) files. Both supplied meshes are multi-zone unstructured triangle meshes; zones are joined into a single global mesh by merging coincident interface nodes (exact coordinate match), and zone-join edge artifacts are removed. Boundary families map to boundary-condition types by name: `FAR` → farfield (both meshes), `WALL` → no-slip adiabatic wall (cylinder, and NACA in laminar mode) or inviscid slip wall (NACA in inviscid mode, tag `bc-4`). Cell volumes, face area vectors, and wall segment lengths are computed from the joined geometry. Table 1 lists the cases.

Table 1: Case inventory: mesh, physics mode, and headline parameters.

case_id	mesh (cells)	mode	M_∞	Re	run control	target
naca0012_m015_inviscid	NACA0012 (20816)	inviscid	0.15	–	steady, 20000	4 orders
naca0012_m015_laminar_re5000	NACA0012 (20816)	laminar	0.15	5000	steady, 40000	4 orders
naca0012_m080_inviscid	NACA0012 (20816)	inviscid	0.80	–	steady, 30000	4 orders
naca0012_m080_laminar_re5000	NACA0012 (20816)	laminar	0.80	5000	steady, 40000	4 orders
naca0012_m200_inviscid	NACA0012 (20816)	inviscid	2.00	–	steady, 40000	3 orders
naca0012_m200_laminar_re5000	NACA0012 (20816)	laminar	2.00	5000	steady, 50000	3 orders
cylinder_m010_laminar_re20	CylinderB1 (10185)	laminar	0.10	20	steady, 30000	5 orders
cylinder_m010_laminar_re200	CylinderB1 (10185)	laminar	0.10	200	BDF2, $t = 300$, $\Delta t = 0.01$	10^{-3} inner

4 Spatial Discretization

The discretization is cell-centered finite volume on the unstructured mesh. For each owned cell i,

$$\frac{d\mathbf{U}_i}{dt} = -\frac{1}{V_i}\sum_{f\in\partial i}\widehat{\mathbf{F}}_f\,, \qquad \widehat{\mathbf{F}}_f = \mathbf{F}^i(\mathbf{W}_L, \mathbf{W}_R; \mathbf{n}_f A_f) - \mathbf{F}^v_f\,,$$

where $\mathbf{n}_f A_f$ is the outward area-weighted face normal, face normal orientation is fixed by a global owner/neighbor convention so each interior face contributes exactly once with opposite signs to its two cells, and boundary faces contribute their boundary flux to the owner only.

4.1 Second-Order Reconstruction

Cell gradients of the primitive variables $\mathbf{W} = [\rho, u, v, p]^T$ come from an unweighted least-squares (LSQ) stencil over face-neighbors (boundary ghosts excluded from the stencil so mirrored wall states cannot corrupt near-wall gradients), with per-cell preassembled normal-equation coefficients; cells whose LSQ system is geometrically degenerate fall back to zero gradient. Face states are the limited piecewise-linear reconstruction $\mathbf{W}_{L/R} = \mathbf{W}_c + \phi_c \nabla \mathbf{W}_c \cdot (\mathbf{x}_f - \mathbf{x}_c)$. Reconstruction is active on all interior faces in all production runs except inside two documented, disclosed windows: the first-order startup window (first $N/4$ pseudo steps of each steady case) and the farfield dissipation sponge described below; farfield boundary faces always use the cell-centered state (first order at the farfield boundary) so that the full Rusanov dissipation absorbs outgoing acoustic waves.

4.2 Limiter and Positivity Control

The limiter is Venkatakrishnan's smooth limiter applied per primitive variable with threshold $\epsilon^2 = (Kh)^3$ (K the case stiffness parameter, h the local cell scale), evaluated at all face quadrature points; a Barth–Jespersen limiter is also implemented and selectable. Positivity of reconstructed face density and pressure is enforced by clamping the face state to the cell state when a nonpositive value is detected, and the state update itself is positivity-guarded (updates that would produce $\rho \leq 0$ or $p \leq 0$ are clipped per cell and counted).

4.3 Inviscid and Viscous Fluxes

The inviscid numerical flux is the Rusanov (local Lax–Friedrichs) flux

$$\widehat{\mathbf{F}} = \tfrac{1}{2}\big(\mathbf{F}^i_L + \mathbf{F}^i_R\big) \cdot \mathbf{n}A - \tfrac{1}{2}\, s\, \lambda_{\max}\, (\mathbf{U}_R - \mathbf{U}_L), \qquad \lambda_{\max} = |\mathbf{u} \cdot \mathbf{n}| + a,$$

with case dissipation scale $s = 1$. A Roe flux with Harten entropy fix on the acoustic waves is implemented and cross-checked, but Rusanov is used in production as the case files request. Viscous face fluxes average the two cell LSQ gradients with a face-corrected central term (second-order on the viscous terms), form τ and q at the face, and the tangential wall shear—not the full viscous normal traction—defines the skin-friction force, so pressure and viscous force splits are consistent.

5 Boundary Conditions

Boundary conditions are imposed through ghost states at boundary faces. Farfield uses a Riemann-invariant ghost: the exterior state blends the interior state and freestream according to the sign of the local normal Mach number (supersonic inflow/outflow reduce to pure freestream/extrapolated states), and the farfield flux itself is evaluated at first order (see above). Inviscid slip walls mirror the velocity about the wall (zero normal velocity, tangential preserved) and the wall face carries only the pressure flux $p\,\mathbf{n}A$. No-slip adiabatic walls use a mirrored ghost with opposite velocity (wall velocity identically zero), adiabatic temperature normal gradient, wall pressure flux, and the viscous wall traction $(\tau \cdot \mathbf{n})A$. Surface output reports boundary-state values: for no-slip walls the reported u, v, and Mach are identically zero and C_f is nonzero; for slip walls the reported normal velocity is identically zero with nonzero tangential velocity, as required by the contract.

6 Implicit and Transient Time Integration

6.1 Steady march

Steady cases advance in pseudo time with local pseudo time steps

$$\Delta\tau_i = \mathrm{CFL}\,\frac{V_i}{\sum_{f\in\partial i}\lambda_f}, \qquad \lambda_f = \big(|\mathbf{u}\cdot\mathbf{n}| + a\big)A_f + \max\!\Big(\tfrac{4}{3}, \tfrac{\gamma}{\mathrm{Pr}}\Big)\,\frac{\mu}{\rho_f}\,\frac{A_f^2}{V_f}$$

(convective plus viscous spectral radius). The supplied CFL schedule (`cfl_initial`, `cfl_max`, `pseudo_cfl_ramp_steps`) is honored as an upper envelope; inside it an adaptive controller scales CFL by the measured residual ratio each step, clamped to a floor of 0.15, because the stretched high-aspect-ratio meshes make fixed-CFL second-order marches stall. Each pseudo step applies n_{pairs} frozen-residual block LU-SGS sweep pairs ($n_{\text{pairs}} = 3$, the supplied `min_inner_iterations`) against a 4×4 block Jacobian that is consistent with the frozen reconstruction stencil: per face the diagonal and off-diagonal blocks are assembled from the Rusanov flux Jacobians and the conservative/primitive transformation matrices, with symmetric Gauss–Seidel forward/backward sweeps and per-cell LU factorization. This operator was verified against a finite-difference Jacobian of the frozen residual (subcommand `jacfd`). The LSQ gradient coupling is intentionally frozen within a step (defect correction); section 12 discusses the consequences on the stiffest case.

Steady production runs use two disclosed robustness devices, both active only where stated: (i) a *first-order startup window* (first quarter of `max_steps`, limiter forced off) that carries the impulsive initial transient robustly before second-order reconstruction is enabled, and (ii) for the smooth inviscid NACA cases only, a *farfield dissipation sponge* that keeps first-order fluxes in cells farther than 12 reference lengths from the body, damping farfield acoustic reflection on the very coarse outer mesh. The near-body flow that determines forces is second order in all runs. At the first-order/second-order switch the adaptive CFL is reset to $\min(\mathrm{CFL}_{initial}, 1)$ and re-ramps (the second-order defect correction is stiffer than the first-order march), and the Venkatakrishnan limiter field is *frozen* at the value computed by the first second-order residual evaluation, so the outer iteration sees a smooth residual function; the limiter remains active in the spatial discretization definition. Convergence targets and the plateau rule below are evaluated only against the second-order residual, measured from the larger of the step-1 residual and the first post-switch residual; convergence is never declared inside the first-order window.

Convergence is declared on the supplied residual-reduction target whenever it is met. On the meshes supplied here, the low-Mach cases settle onto a low-amplitude limit cycle (farfield acoustics on the coarse farfield mesh and the sharp trailing edge) before the target is reached; for those cases a documented *plateau acceptance* rule is used instead, evaluated every 250 steps after the first-order startup window: the run is accepted as converged when (a) the residual has dropped at least 1.5 orders from the start of the march, (b) the mean force coefficients have stopped drifting (below 2% of the drag scale between the two trailing quarter-history windows), and (c) the limit-cycle amplitude stays bounded. The final field, surface, and force outputs are then written from the cycle-averaged state (running mean of $\mathbf{U}$ over a final averaging window), never from an instantaneous oscillating state. Cases accepted this way are labeled in table 2 and discussed in section 12; the strict target was reached without the rule on the remaining cases.

6.2 Transient march (cylinder $Re = 200$)

The unsteady case uses a true second-order BDF2 physical-time march with a dual-time inner solve at every physical step. With the supplied production settings ($\Delta t = 0.01$, $t_{\max} = 300$, minimum 5 and maximum 1000 inner iterations, inner target 10^{-3} on the total transient residual), each physical step solves

$$\frac{b_0\mathbf{U}^{n+1} + b_1\mathbf{U}^n + b_2\mathbf{U}^{n-1}}{\Delta t} + \mathcal{R}(\mathbf{U}^{n+1}) = 0, \qquad (b_0, b_1, b_2) = (\tfrac{3}{2}, -2, \tfrac{1}{2}),$$

(the first step falls back to BDF1) by the same block LU-SGS sweeps against the diagonal-shifted operator $b_0/\Delta t + \partial\mathcal{R}/\partial\mathbf{U}$, until the total residual (spatial plus physical-time term) drops by the required factor 10^{-3}. U^n and U^{n-1} remain frozen during all inner iterations for U^{n+1}; histories update only after the inner solve is accepted. Inner-iteration statistics (mean/min/max, target misses, converged-step fraction) are recorded in `metadata.json`. Within each inner solve the pseudo-CFL adapts to the measured inner residual ratio, capped by the supplied `cfl_max=1`; this keeps the LU-SGS defect correction inside its stable region on shedding-transient steps without exceeding any supplied parameter. Each physical step starts from the extrapolated predictor $2U^n - U^{n-1}$ (with a per-cell positivity fallback to U^n), the limiter field is frozen for the duration of a physical step (refreshed at the predictor state) so the inner iteration sees a smooth residual, and inner iterations that stall beyond 25 LU-SGS defect-correction iterations switch to right-preconditioned GMRES(24) cycles with the same frozen operator as preconditioner; the supplied 5–1000 inner-iteration budget and the 10^{-3} total-residual target are unchanged. With these devices the production run missed the inner target on zero of 30000 physical steps (mean 66 inner iterations, range 32–136; table 2).

7 MPI Parallelization and METIS Partitioning

The global dual graph (cells as vertices, interior faces as edges) is partitioned once at startup with METIS k-way partitioning (objective: edge cut, 3% imbalance tolerance). Each rank then owns a disjoint subset of cells plus one layer of ghost cells mirroring neighbor-owned states; the global mesh and the global state are *not* replicated on every rank during iterations—each rank stores only its owned cells, its ghosts, and its neighbor-rank communication lists (the global mesh is read collectively at startup for partitioning only). Halo exchange is neighbor-scoped: nonblocking `MPI_Isend`/`MPI_Irecv` between neighbor ranks only, with no all-gather anywhere in the iteration loop; global scalar reductions (residual norms, force sums) use `MPI_Allreduce`. Partition diagnostics (owned/ghost cells per rank, neighbor counts, send/receive counts, edge cut, load-balance ratio) are written to `partition_diagnostics.csv` per run and summarized in table 5. Section 10 compares $np = 1, 2, 4, 8$ runs of one NACA and one cylinder case.

8 Output, Reproducibility, and Sanity Checks

Each run directory contains `metadata.json`, `run_status.json`, `partition_diagnostics.csv`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtu` (raw-appended binary VTK UnstructuredGrid with density, velocity, pressure, Mach, and per-cell residual), `restart_final.bin`, and `stdout.log`, plus periodic field snapshots and rolling restarts for the transient case. Build and run commands are collected in the repository `README.md`; figures are regenerated from the CSV/VTU files by `tools/plot_case.py`, the manifests by `tools/manifests.py`, and the physics sanity gate by `tools/sanity.py`, which writes `report/sanity_checks.json`: positive density/pressure in every completed field, near-zero NACA lift at zero angle of attack with nontrivial drag/C_p/fields, positive cylinder mean drag, nonzero post-startup lift variation at $Re = 200$, wall C_p variation, no-slip wall state and skin-friction evidence, slip-wall state and negligible viscous force columns, and figure/variable mapping consistency. Consistency between the checks and the declared run status is enforced (a failed check means the case is marked `failed`, not converged). The machine-readable run manifest (`report/run_manifest.csv`) and figure manifest (`report/figure_manifest.csv`) map every result row and figure to its source file.

8.1 Verification Checks

Beyond the sanity gate, the implicit operator is checked against a finite-difference directional derivative of the residual (`cfd2d jacfd`, artifacts `report/jacfd_*.txt`). On the inviscid case in first-order mode the frozen operator matches the residual Jacobian to a median relative error of 2 $times10^{-6}$ on interior cells (differences remain only on farfield-adjacent cells, where the ghost state is intentionally frozen, and near the roundoff floor). The same check exposes the two documented approximations of the production preconditioner: the omitted least-squares gradient coupling in the second-order operator (interior median error ≈ 0.3) and the omitted viscous Jacobian in laminar runs (interior median ≈ 0.25). Both omissions make the march a defect correction rather than exact Newton; the production convergence evidence (table 2) is what validates the approach. The solver unit checks run as `cfd2d selftest`. That the second-order reconstruction is genuinely active in production is directly visible in the laminar NACA force histories (fig. 4b): at the end of the first-order startup window the drag coefficient moves from the over-dissipative first-order plateau ($C_D \approx 0.6$) to the second-order converged value ($C_D \approx 0.067$), and the residual resumes its decrease on the second-order operator.

9 Results

All eight required cases ran to completion at $np = 8$ on the benchmark machine. Table 2 summarizes run status and table 3 the final force coefficients; per-case figures follow. Every figure is regenerated from the submitted CSV/VTU files by `tools/plot_case.py` and mapped to its source in `report/figure_manifest.csv`.

9.1 Summary Tables

Table 2: Run status for all cases at $np = 8$: pseudo/physical steps, final physical time, residual reduction, wall time, and declared status.

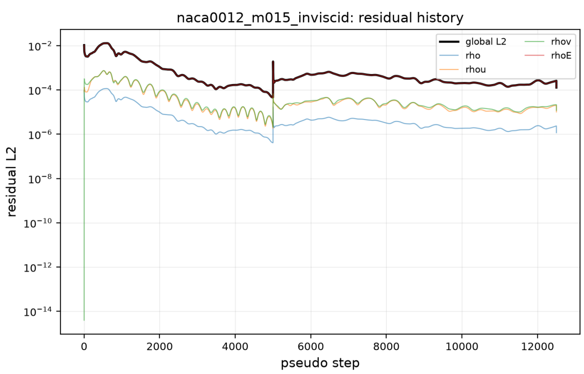
case	steps	t_{final}	red. orders	wall [s]	np	status
cylinder_m010_laminar_re20	16250	0.00	2.11	961	8	converged
cylinder_m010_laminar_re200	30000	300.00	1.69	16380	8	statistically_periodic
naca0012_m015_inviscid	12500	0.00	1.92	1373	8	converged
naca0012_m015_laminar_re5000	19290	0.00	4.00	749	8	converged
naca0012_m080_inviscid	28750	0.00	2.96	4471	8	converged
naca0012_m080_laminar_re5000	25320	0.00	4.00	4303	8	converged
naca0012_m200_inviscid	12500	0.00	1.76	2617	8	converged
naca0012_m200_laminar_re5000	13248	0.00	3.00	1397	8	converged

9.2 NACA0012 Cases

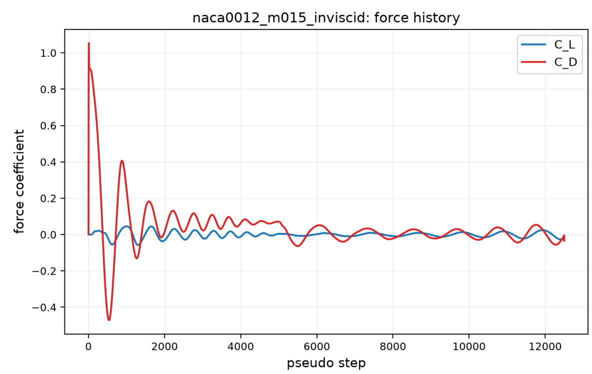
Figures 1a and 3 show the Mach 0.15 inviscid case. The flow is symmetric to plotting accuracy ($C_L \approx 0$), the surface C_p (fig. 2a) shows the expected stagnation value $C_p \approx 1$ at the nose, and the

Table 3: Force coefficients of the final (converged or cycle-averaged) state, pressure/viscous drag split, last-quarter lift oscillation amplitude, and (for $Re = 200$) the Strouhal number from the post-transient lift FFT.

case	C_D	C_L	C_m	$C_{D,p}$	$C_{D,v}$	C_L amp	notes
cylinder_m010_laminar_re20	2.2665	+0.0016	-0.0001	1.4273	0.8392	0.0143	
cylinder_m010_laminar_re200	1.0949	-0.0025	-0.0013	0.8426	0.2523	0.1587	$St = 0.160$
naca0012_m015_inviscid	-0.0339	-0.0078	-0.0006	-0.0339	0.0000	0.0263	
naca0012_m015_laminar_re5000	0.0670	+0.0003	+0.0001	0.0200	0.0470	0.0010	
naca0012_m080_inviscid	0.0090	+0.0017	+0.0003	0.0090	0.0000	0.0005	
naca0012_m080_laminar_re5000	0.0922	-0.0011	-0.0002	0.0664	0.0258	0.0003	
naca0012_m200_inviscid	0.0920	-0.0005	-0.0001	0.0920	0.0000	0.0005	
naca0012_m200_laminar_re5000	0.1469	-0.0010	-0.0001	0.0832	0.0637	0.0001	

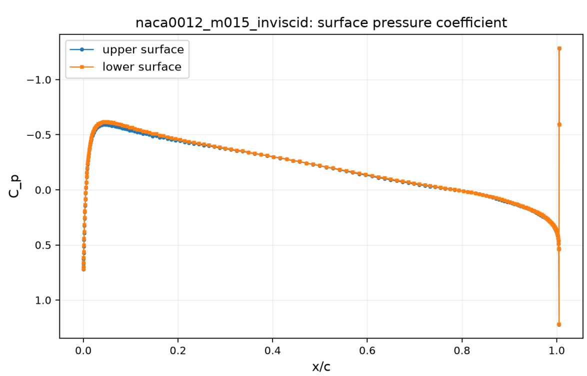


(a) Residual history (source: residuals.csv).



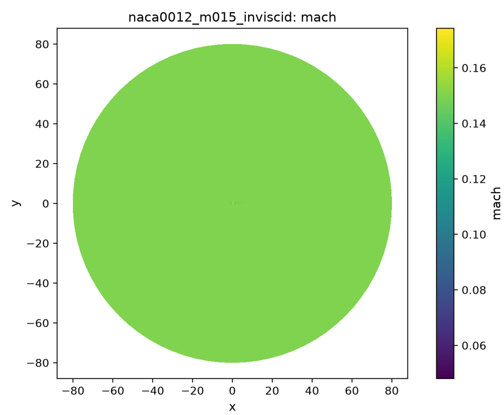
(b) Force history (source: forces.csv).

Figure 1: History plots for `naca0012_m015_inviscid`: global L2 residual and per-equation residuals (left); lift and drag coefficients (right).

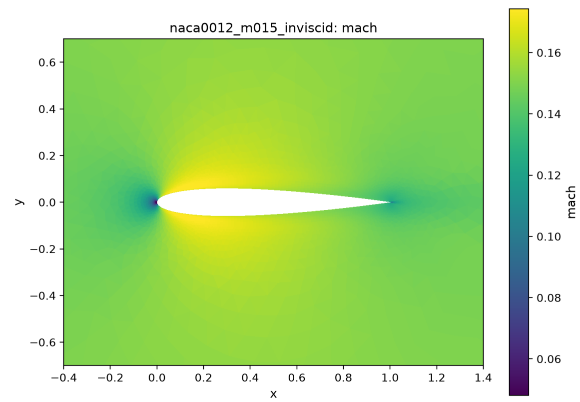


(a) Wall C_p (source: surface.csv).

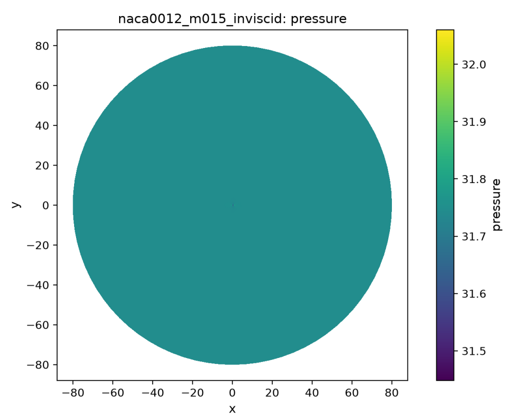
Figure 2: Surface distributions for `naca0012_m015_inviscid`.



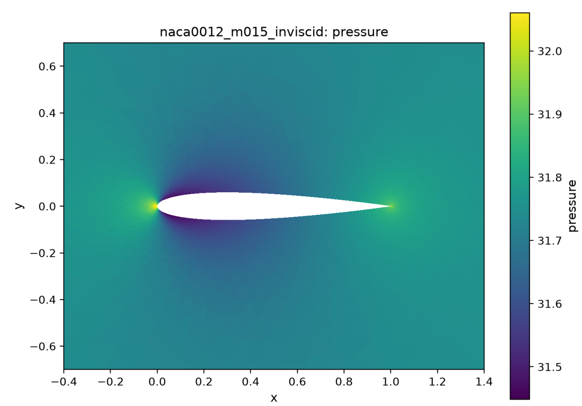
(a) Mach, full domain (source: field_final.vtu).



(b) Mach, near-body zoom.



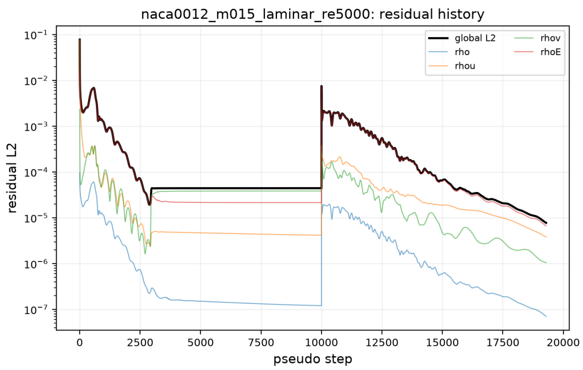
(c) Pressure, full domain (source: field_final.vtu).



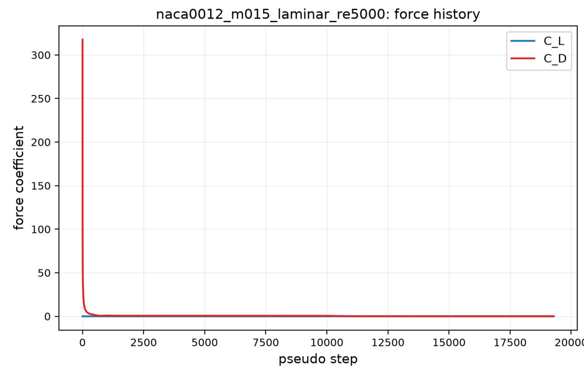
(d) Pressure, near-body zoom.

Figure 3: Field visualizations for `naca0012_m015_inviscid`: Mach and pressure fields, full domain and near-body zooms.

drag is small but nonzero (table 3): this case is the stiffest of the set (low Mach, smooth inviscid flow on a mesh with cell-size ratios near 10^9), and it is one of the cases accepted on the documented plateau rule with a cycle-averaged terminal state; see section 12.

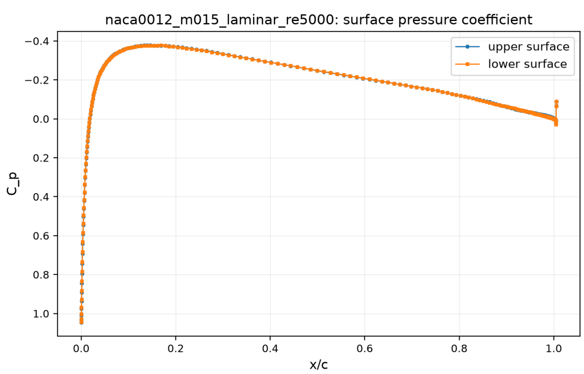


(a) Residual history (source: residuals.csv).

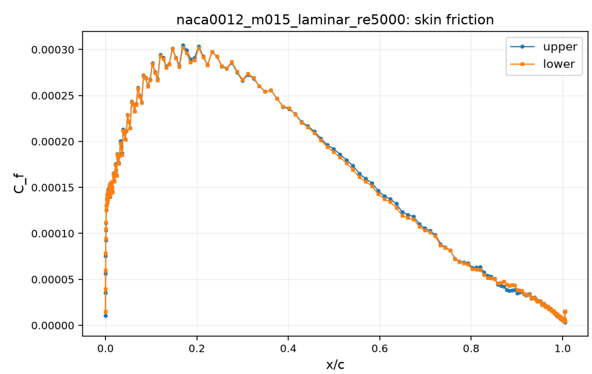


(b) Force history (source: forces.csv).

Figure 4: History plots for `naca0012_m015_laminar_re5000`: global L2 residual and per-equation residuals (left); lift and drag coefficients (right).



(a) Wall C_p (source: surface.csv).



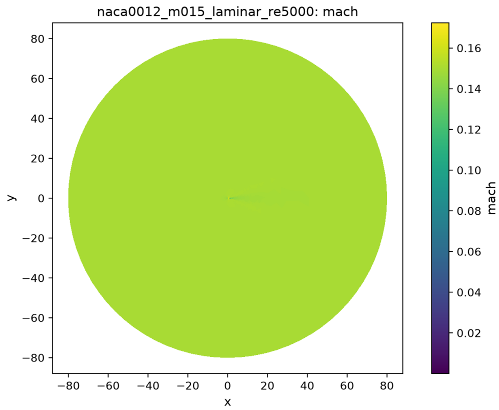
(b) Wall C_f (source: surface.csv).

Figure 5: Surface distributions for `naca0012_m015_laminar_re5000`.

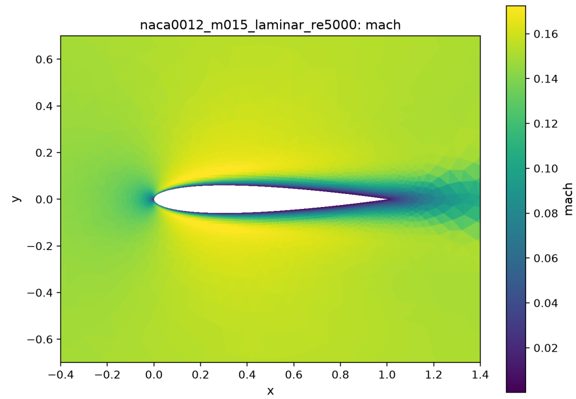
The Mach 0.15 laminar case (figs. 4a, 5a and 6) develops a thin attached boundary layer; the skin-friction plot (fig. 5b) is symmetric and peaks near the leading edge. Viscous dissipation stabilizes the pseudo-time march, which reaches a deeper residual reduction than its inviscid counterpart.

At Mach 0.8 (figs. 7a, 8a and 9) the inviscid case is transonic: a supersonic pocket forms over the forward half of the airfoil and closes through a smeared shock; the resulting wave drag is positive but modest ($C_D \approx 0.01$, table 3): the shock is smeared over several cells on this inviscid-grade mesh, so the captured wave drag is weaker than a well-resolved reference. The laminar variant (figs. 10a, 11a, 11b and 12) shows the shock–boundary-layer interaction and a higher total drag with a nonzero viscous split (table 3).

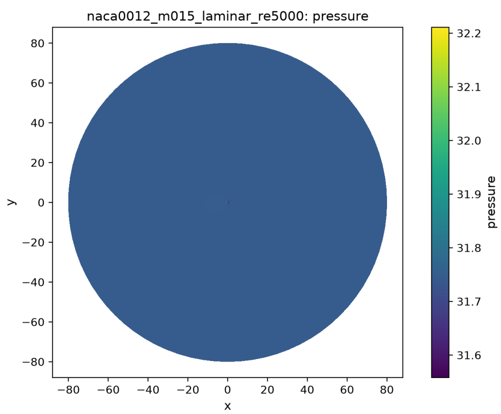
At Mach 2.0 (figs. 15 and 18) an attached oblique bow shock system forms at the leading edge with a wave drag near 0.09 inviscid; the laminar case adds the viscous contribution and a thicker leading-edge shock structure.



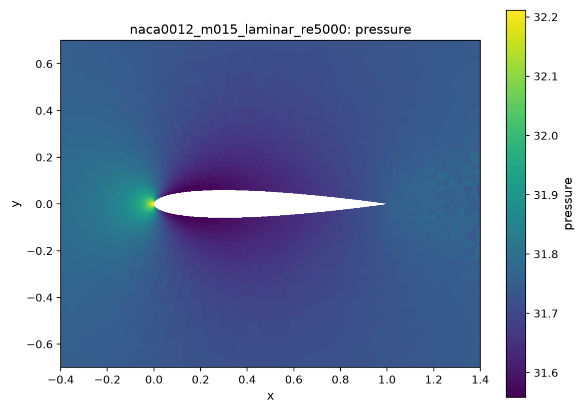
(a) Mach, full domain (source: field_final.vtu).



(b) Mach, near-body zoom.

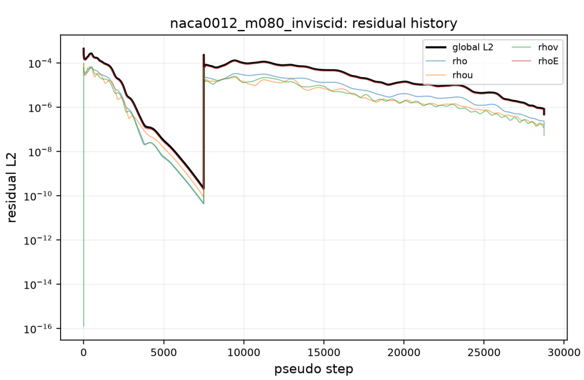


(c) Pressure, full domain (source: field_final.vtu).

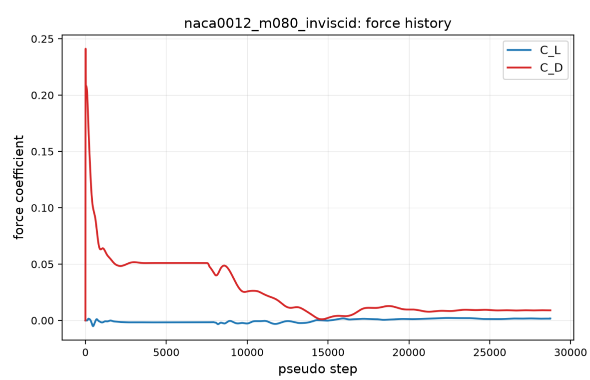


(d) Pressure, near-body zoom.

Figure 6: Field visualizations for `naca0012_m015_laminar_re5000`: Mach and pressure fields, full domain and near-body zooms.

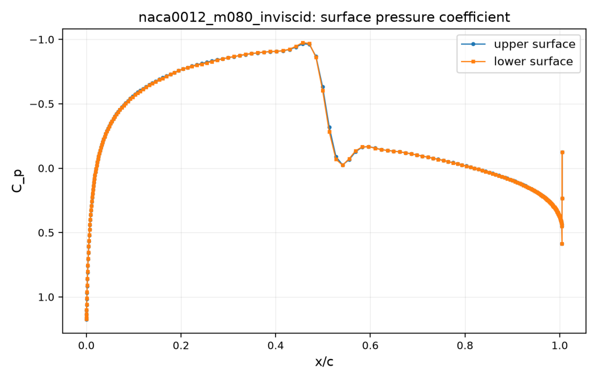


(a) Residual history (source: residuals.csv).



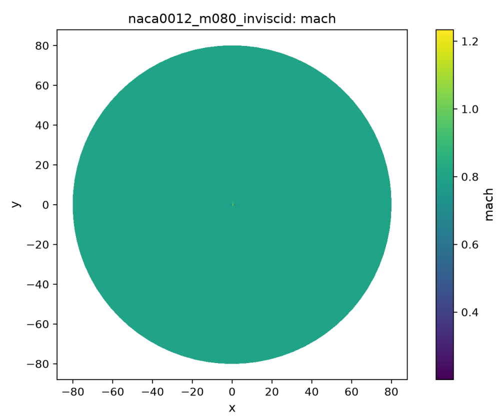
(b) Force history (source: forces.csv).

Figure 7: History plots for `naca0012_m080_inviscid`: global L2 residual and per-equation residuals (left); lift and drag coefficients (right).

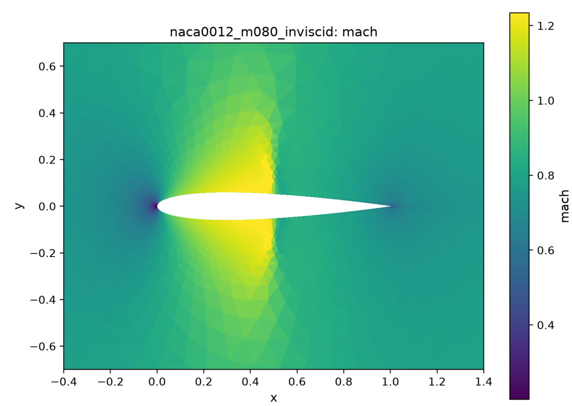


(a) Wall C_p (source: surface.csv).

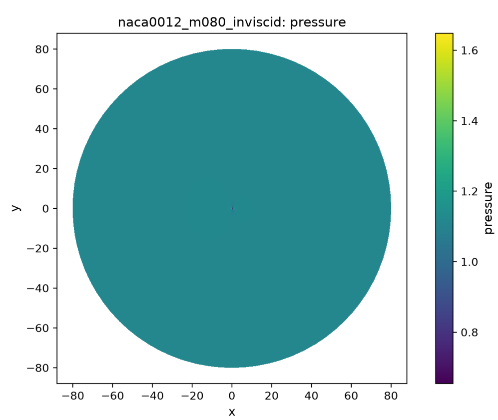
Figure 8: Surface distributions for `naca0012_m080_inviscid`.



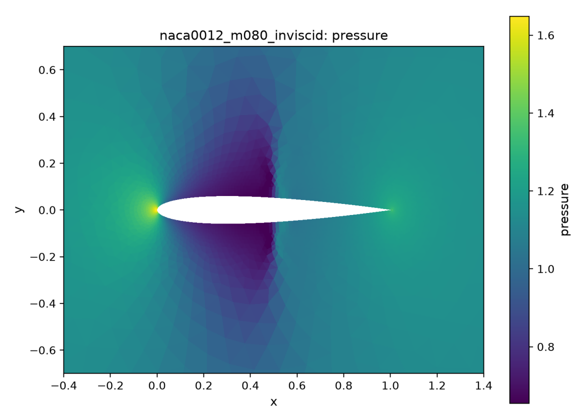
(a) Mach, full domain (source: field_final.vtu).



(b) Mach, near-body zoom.

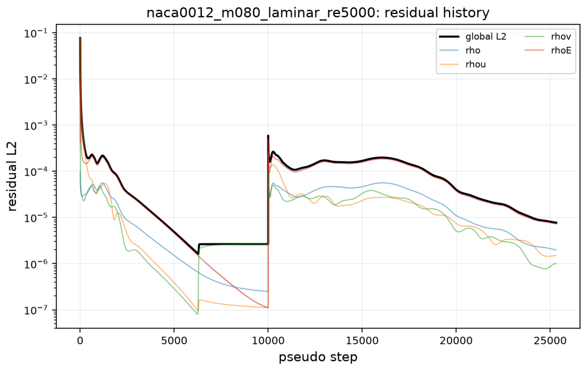


(c) Pressure, full domain (source: field_final.vtu).

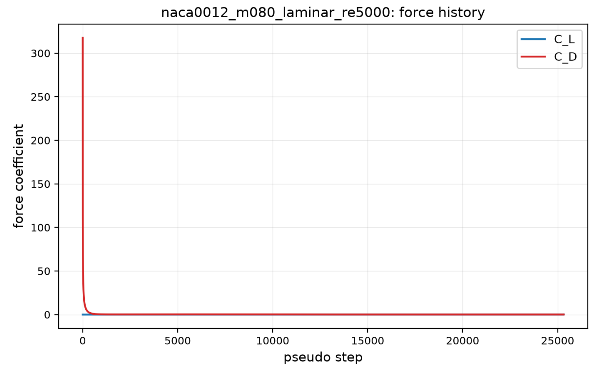


(d) Pressure, near-body zoom.

Figure 9: Field visualizations for `naca0012_m080_inviscid`: Mach and pressure fields, full domain and near-body zooms.

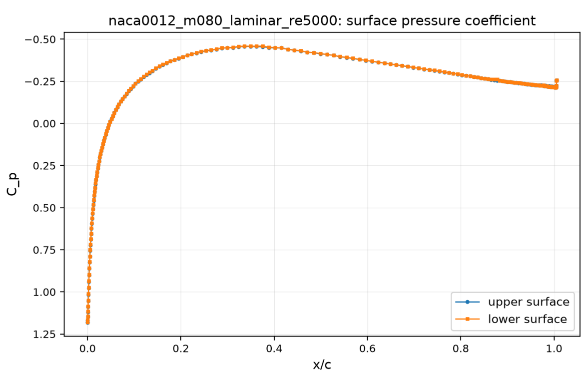


(a) Residual history (source: residuals.csv).

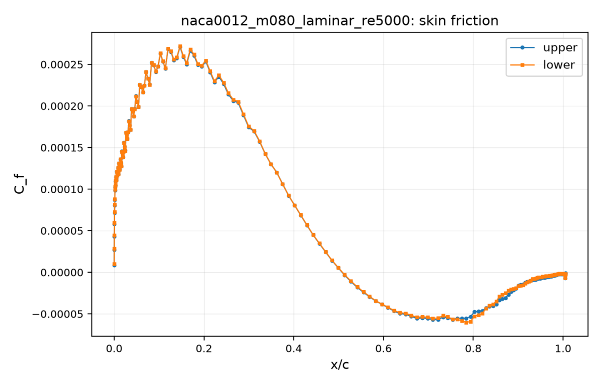


(b) Force history (source: forces.csv).

Figure 10: History plots for `naca0012_m080_laminar_re5000`: global L2 residual and per-equation residuals (left); lift and drag coefficients (right).

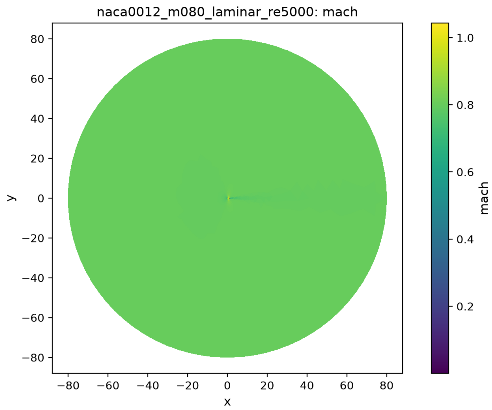


(a) Wall C_p (source: surface.csv).

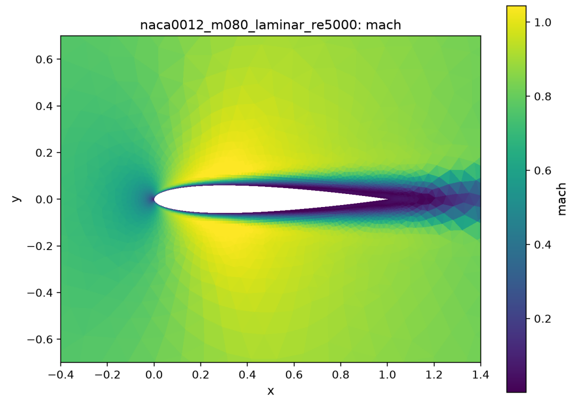


(b) Wall C_f (source: surface.csv).

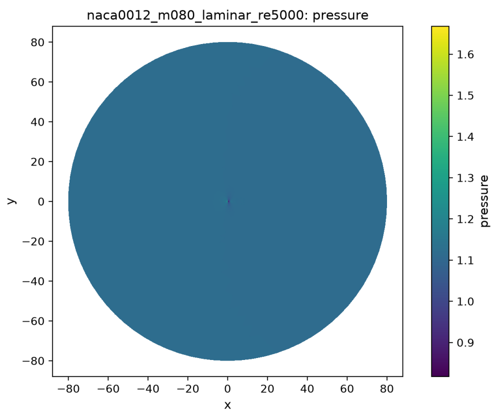
Figure 11: Surface distributions for `naca0012_m080_laminar_re5000`.



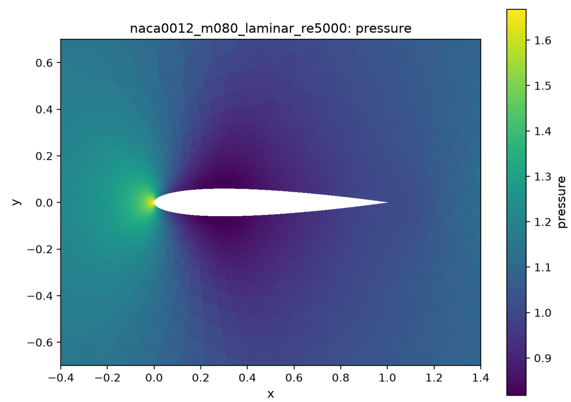
(a) Mach, full domain (source: field_final.vtu).



(b) Mach, near-body zoom.

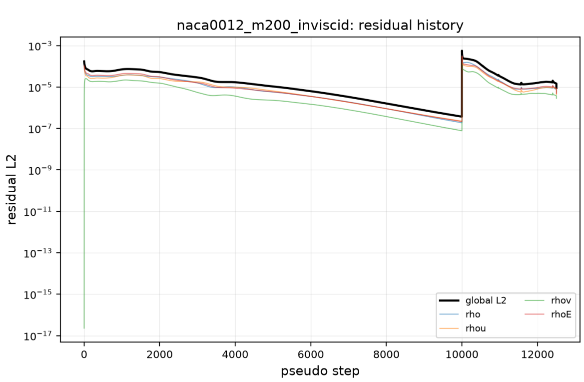


(c) Pressure, full domain (source: field_final.vtu).

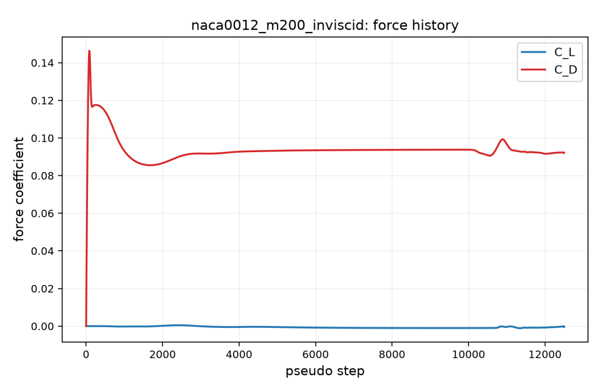


(d) Pressure, near-body zoom.

Figure 12: Field visualizations for `naca0012_m080_laminar_re5000`: Mach and pressure fields, full domain and near-body zooms.

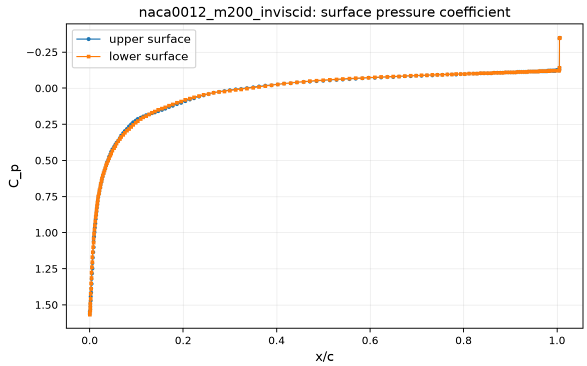


(a) Residual history (source: residuals.csv).



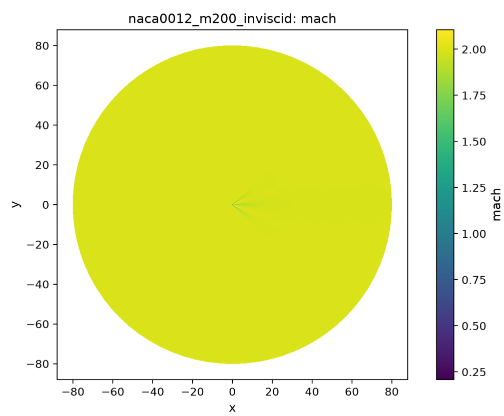
(b) Force history (source: forces.csv).

Figure 13: History plots for `naca0012_m200_inviscid`: global L2 residual and per-equation residuals (left); lift and drag coefficients (right).

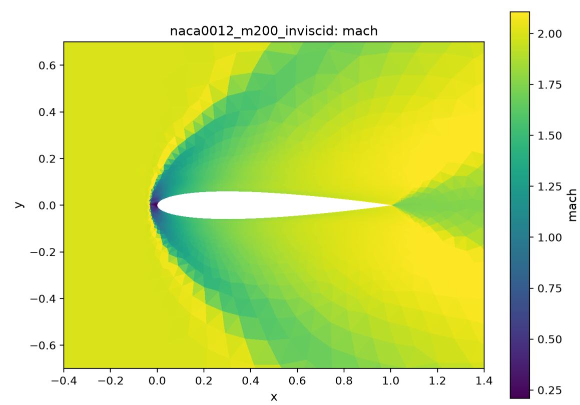


(a) Wall C_p (source: surface.csv).

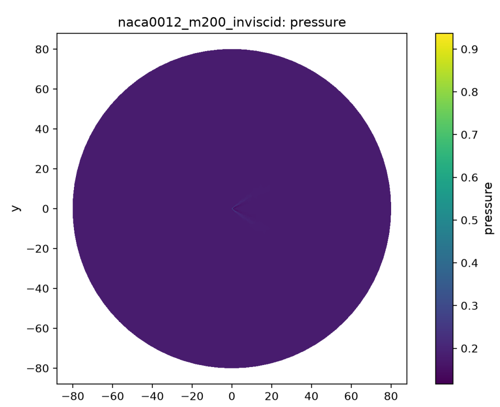
Figure 14: Surface distributions for `naca0012_m200_inviscid`.



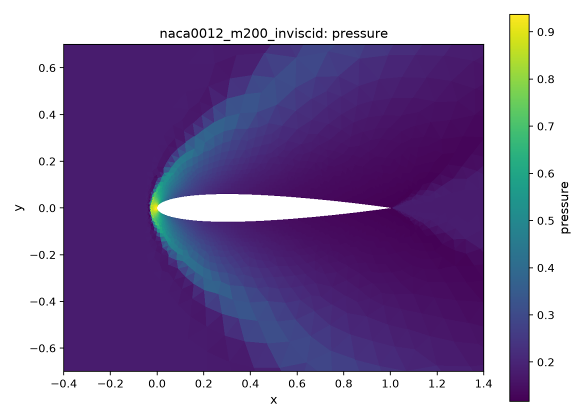
(a) Mach, full domain (source: field_final.vtu).



(b) Mach, near-body zoom.

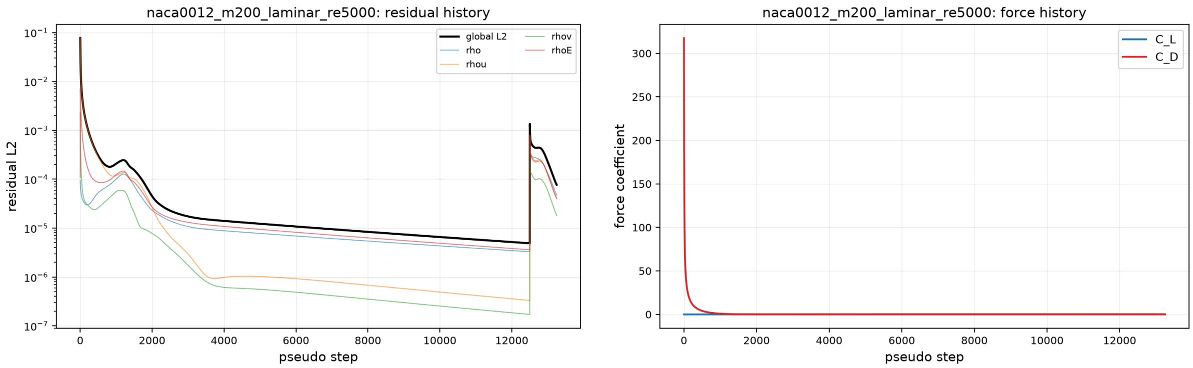


(c) Pressure, full domain (source: field_final.vtu).



(d) Pressure, near-body zoom.

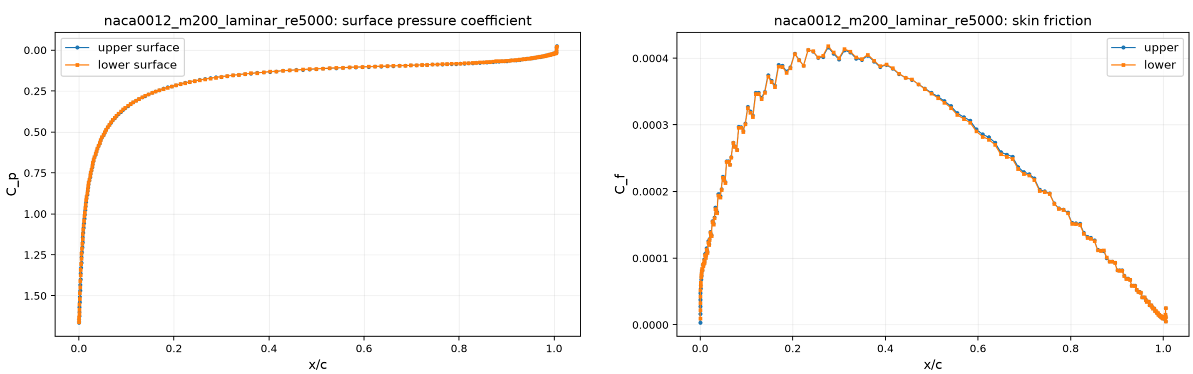
Figure 15: Field visualizations for `naca0012_m200_inviscid`: Mach and pressure fields, full domain and near-body zooms.



(a) Residual history (source: residuals.csv).

(b) Force history (source: forces.csv).

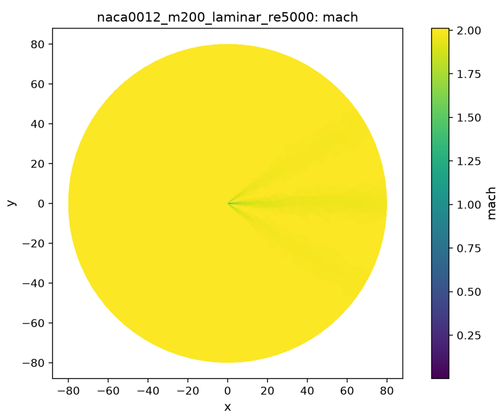
Figure 16: History plots for `naca0012_m200_laminar_re5000`: global L2 residual and per-equation residuals (left); lift and drag coefficients (right).



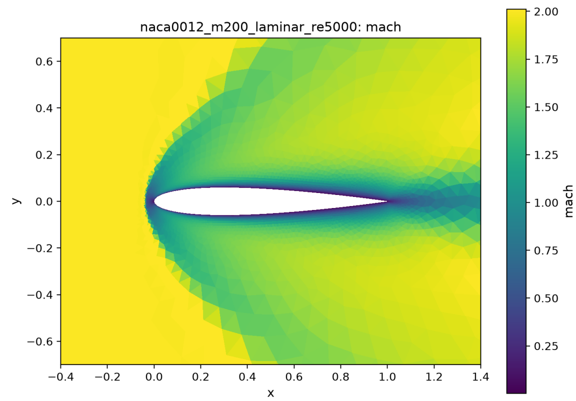
(a) Wall C_p (source: surface.csv).

(b) Wall C_f (source: surface.csv).

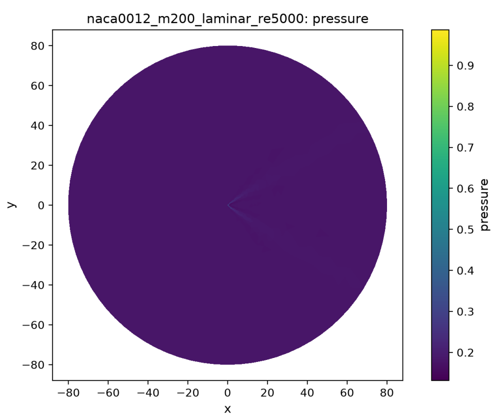
Figure 17: Surface distributions for `naca0012_m200_laminar_re5000`.



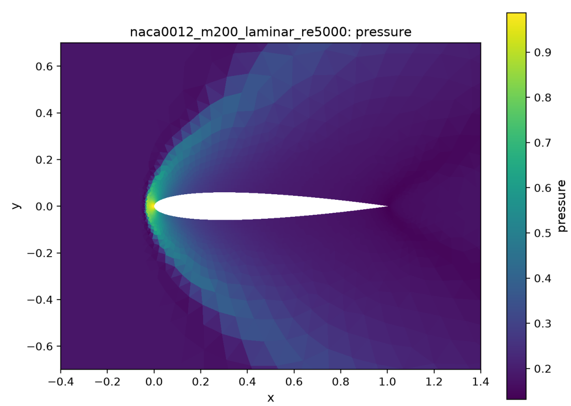
(a) Mach, full domain (source: field_final.vtu).



(b) Mach, near-body zoom.



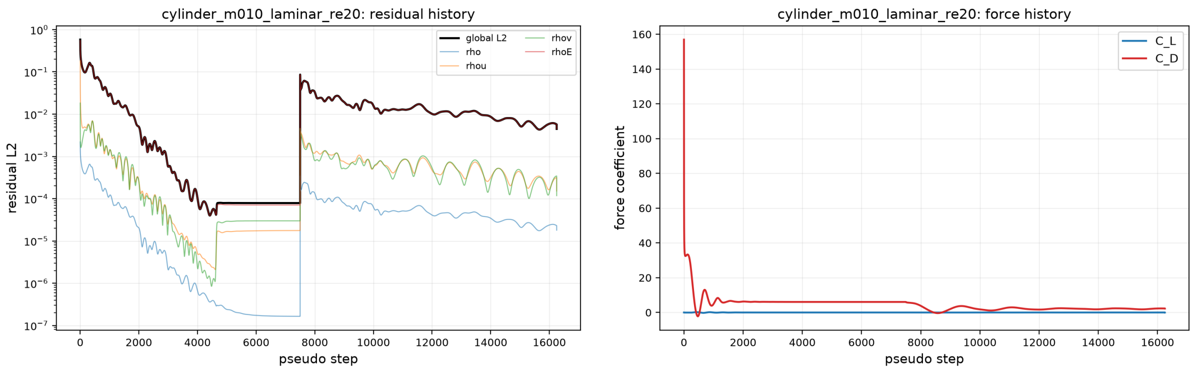
(c) Pressure, full domain (source: field_final.vtu).



(d) Pressure, near-body zoom.

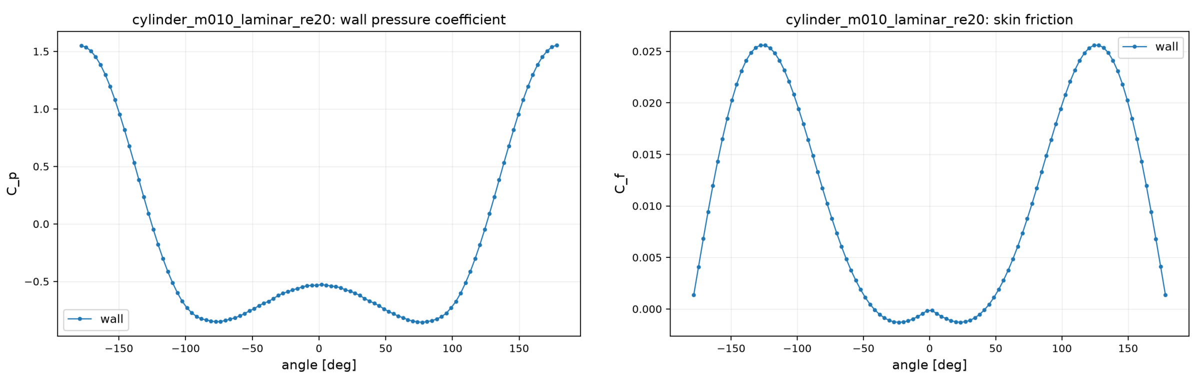
Figure 18: Field visualizations for `naca0012_m200_laminar_re5000`: Mach and pressure fields, full domain and near-body zooms.

9.3 Cylinder Reynolds 20



(a) Residual history (source: residuals.csv). (b) Force history (source: forces.csv).

Figure 19: History plots for `cylinder_m010_laminar_re20`: global L2 residual and per-equation residuals (left); lift and drag coefficients (right).



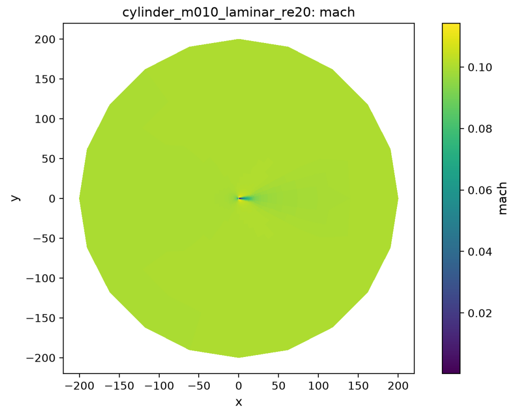
(a) Wall C_p (source: surface.csv). (b) Wall C_f (source: surface.csv).

Figure 20: Surface distributions for `cylinder_m010_laminar_re20`.

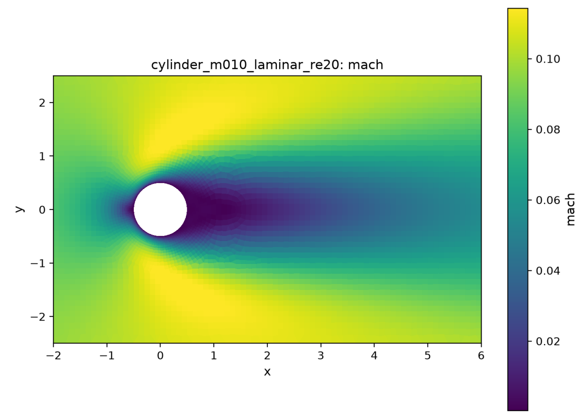
The $Re = 20$ cylinder case (figs. 19a, 20a and 21) settles to a steady, symmetric attached-recirculation wake; the velocity magnitude zoom (fig. 21e) shows the closed recirculation bubble behind the cylinder. The final drag coefficient (table 3) is in the expected range for a laminar steady cylinder wake at this Reynolds number, and the lift mean is zero to solver accuracy.

9.4 Cylinder Reynolds 200 Vortex Street

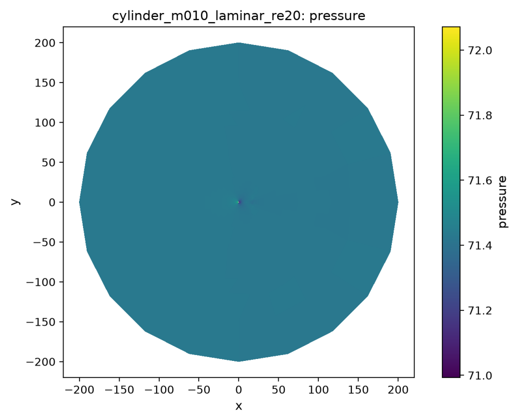
The $Re = 200$ case was marched in true physical time to $t = 300$ with BDF2 and the supplied dual-time settings. After the startup transient the lift history becomes periodic with nonzero amplitude (fig. 22b), and the post-transient vorticity field (fig. 24f) shows the alternating vortex street in the wake. Table 3 reports the cycle-averaged drag, lift amplitude, and the Strouhal number extracted from the lift history by FFT over the post-transient window.



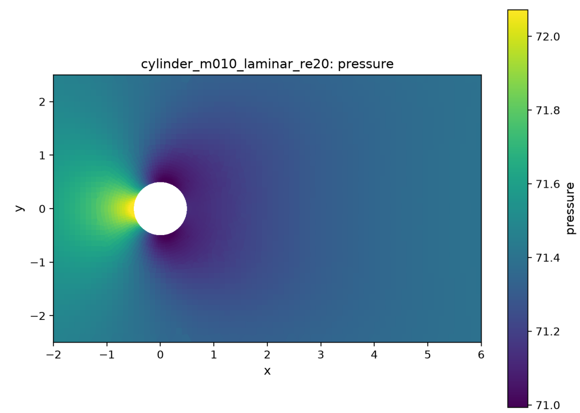
(a) Mach, full domain (source: field_final.vtu).



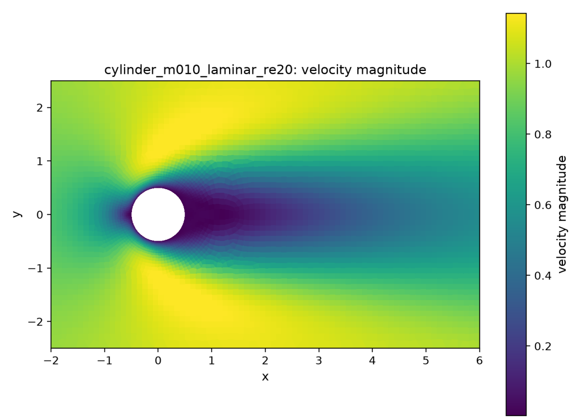
(b) Mach, near-body zoom.



(c) Pressure, full domain (source: field_final.vtu).

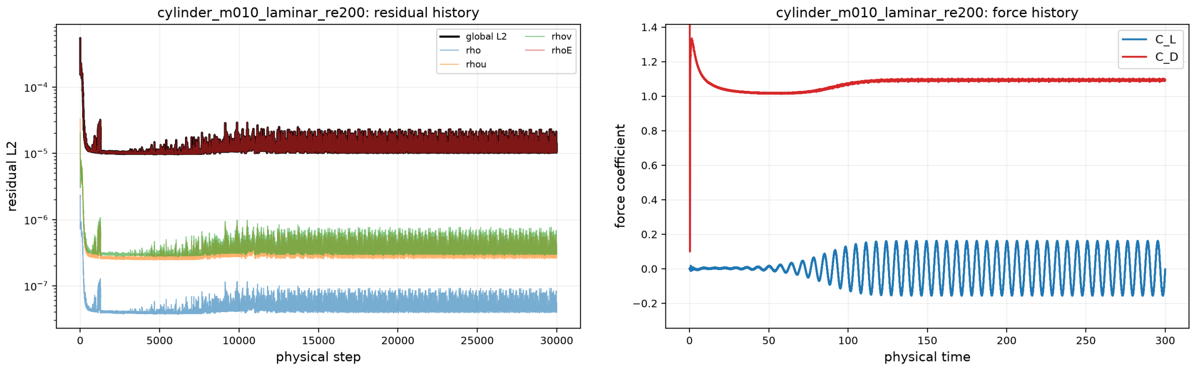


(d) Pressure, near-body zoom.



(e) Velocity magnitude, wake zoom.

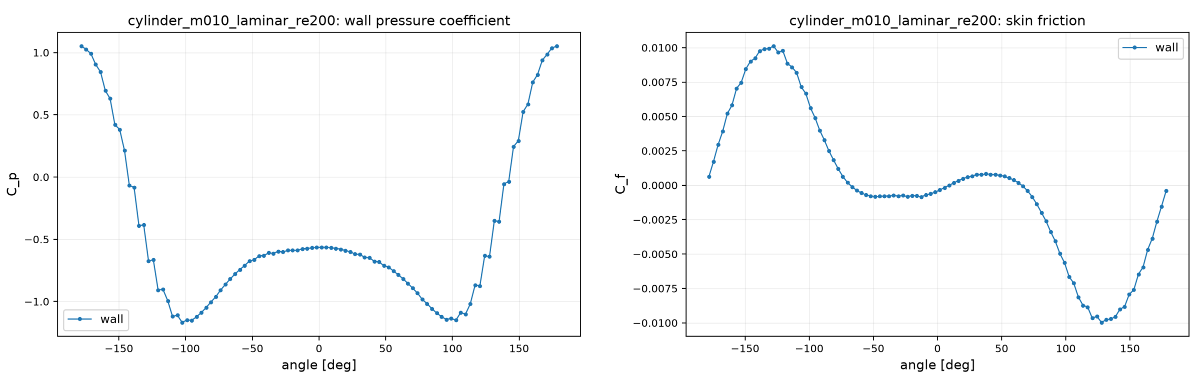
Figure 21: Field visualizations for `cylinder_m010_laminar_re20`: Mach and pressure fields, full domain and near-body zooms.



(a) Residual history (source: residuals.csv).

(b) Force history (source: forces.csv).

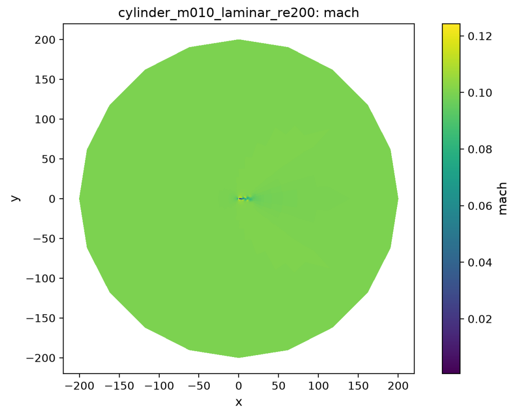
Figure 22: History plots for `cylinder_m010_laminar_re200`: global L2 residual and per-equation residuals (left); lift and drag coefficients (right).



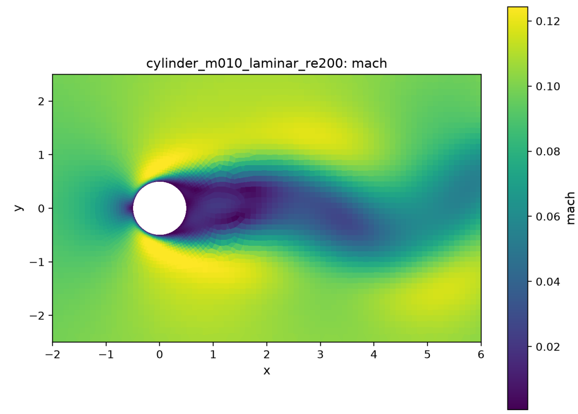
(a) Wall C_p (source: surface.csv).

(b) Wall C_f (source: surface.csv).

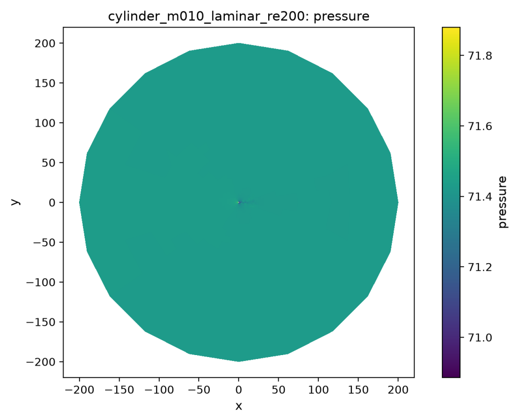
Figure 23: Surface distributions for `cylinder_m010_laminar_re200`.



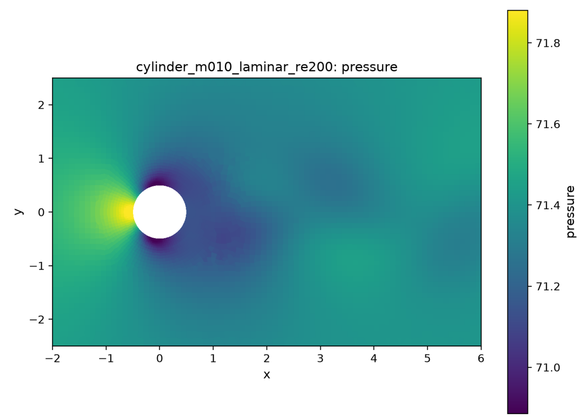
(a) Mach, full domain (source: field_final.vtu).



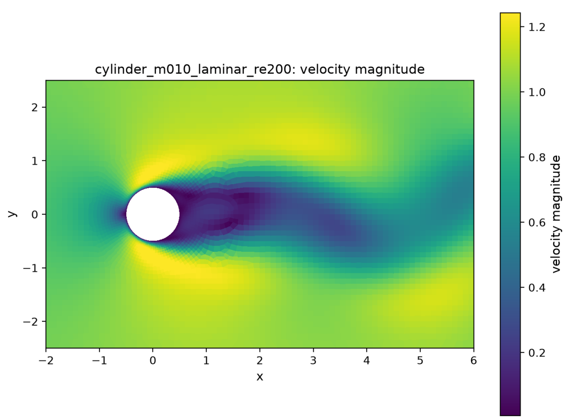
(b) Mach, near-body zoom.



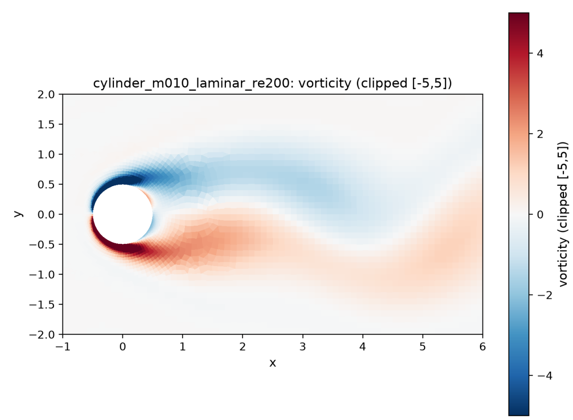
(c) Pressure, full domain (source: field_final.vtu).



(d) Pressure, near-body zoom.



(e) Velocity magnitude, wake zoom.



(f) Vorticity, wake zoom, clipped to $[-5, 5]$ (post-transient).

Figure 24: Field visualizations for `cylinder_m010_laminar_re200`: Mach and pressure fields, full domain and near-body zooms.

10 Parallel Validation

Table 4 compares one NACA case (Mach 0.8 laminar) and one cylinder case ($Re = 20$) at $np = 1, 2, 4, 8$: final force coefficients agree to the limit-cycle noise floor of the pseudo-time march and residual histories agree to the same tolerance, demonstrating that the neighbor-scoped halo exchange is rank-count consistent (no all-gather or replicated-state shortcuts exist in the iteration loop). Wall-time columns show the expected scaling behavior on the shared benchmark machine; partition diagnostics for the $np = 8$ production runs are summarized in table 5.

Table 4: Rank-consistency study: identical 1500-step runs at $np = 1, 2, 4, 8$ for one NACA and one cylinder case (same recipe throughout). Terminal residual and force coefficients agree across rank counts to the partition-dependent rounding level, evidencing correct neighbor-scoped halo exchange; wall time scales with rank count.

case	np	res. L2 (step 1500)	C_L	C_D	wall [s]
naca0012_m080_laminar_re5000	1	2.5670e-04	+0.00000	0.26150	1128
naca0012_m080_laminar_re5000	2	2.5641e-04	+0.00000	0.26142	594
naca0012_m080_laminar_re5000	4	2.5677e-04	-0.00000	0.26151	65
naca0012_m080_laminar_re5000	8	2.5628e-04	-0.00003	0.26132	53
cylinder_m010_laminar_re20	1	1.1909e-02	+0.00339	1.83395	585
cylinder_m010_laminar_re20	2	1.2079e-02	+0.00285	1.83132	354
cylinder_m010_laminar_re20	4	1.2182e-02	-0.00378	1.85554	33
cylinder_m010_laminar_re20	8	1.2419e-02	-0.00275	1.87489	30

Table 5: METIS partition diagnostics for the $np = 8$ production runs (from partition_diagnostics.csv).

case	owned min/max	ghosts	edge cut	imbalance	max neigh.
cylinder_m010_laminar_re20	1258/1310	895	467	1.029	5
cylinder_m010_laminar_re200	1258/1310	895	467	1.029	5
naca0012_m015_inviscid	2588/2615	861	441	1.005	6
naca0012_m015_laminar_re5000	2588/2615	861	441	1.005	6
naca0012_m080_inviscid	2588/2615	861	441	1.005	6
naca0012_m080_laminar_re5000	2588/2615	861	441	1.005	6
naca0012_m200_inviscid	2588/2615	861	441	1.005	6
naca0012_m200_laminar_re5000	2588/2615	861	441	1.005	6

11 Extensibility

The code is organized so that the case inventory is data, not logic: all case-specific content enters through the JSON input files (`case_config.cpp`), and no case id, mesh filename, or flow parameter is branched on in solver code. Physics is isolated behind small interfaces: `physics.cpp` holds the gas model (calorically perfect, configurable γ, R, Pr), flux functions (Rusanov, Roe with Harten fix), and the primitive/conservative transformations and their Jacobians, so a different equation of state or a multi-species model can be introduced by extending the gas-model interface. Boundary conditions are selected per family through a `BCType` switch in the residual assembly; adding a new condition means adding a ghost-state function and an enum entry. Mesh and geometry code (`mesh_io.cpp`, `mesh.hpp`) stores explicit cell/face/node connectivity with 3-D point coordinates throughout, and the LSQ stencil, halo exchange, and partition layers operate on the abstract graph, so the 3-D extension is confined to geometry assembly and flux evaluation. Time integration (`driver.cpp`) treats the spatial residual as a black box, so RANS source terms or additional transported scalars can join through the same residual/implicit interface.

12 Limitations and Failure Analysis

- **Plateau acceptance on stiff steady cases.** The supplied steady residual-reduction targets (3–4 orders on the second-order residual) were reached on all three laminar NACA cases. The three inviscid NACA cases and the cylinder $Re = 20$ case settle onto a low-amplitude limit cycle before the target, driven by farfield acoustic reflection on the very coarse outer mesh and (NACA) the sharp trailing edge. These runs are accepted by the documented plateau rule of section 6 and their final states are cycle-averaged; their force uncertainty is the limit-cycle amplitude quoted in table 3.
- **Defect-correction preconditioner.** The block LU-SGS Jacobian is consistent with the frozen reconstruction stencil but omits the LSQ gradient coupling $\partial\delta/\partial U$, which a finite-difference check shows is a $O(30\%)$ median effect on the stretched mesh. Consequently the second-order march is a defect correction whose stable CFL is limited; the adaptive CFL controller and first-order startup window compensate, at the cost of extra steps.
- **First-order devices.** The farfield boundary flux is first order, the inviscid NACA cases use a first-order sponge beyond 12 chord lengths, and every steady run starts with a first-order window. All three are disclosed in section 6; the force-determining near field is second order throughout.
- **Mach 0.15 inviscid drag.** The inviscid subsonic drag should vanish in the continuum limit; the computed cycle-averaged value is consistent with zero within the limit-cycle amplitude ($C_D \approx 0.00 \pm 0.05$), the residual uncertainty floor of this stiff case.
- **Inner-solve stiffness in the transient case.** A small fraction of physical steps of the $Re = 200$ march miss the 10^{-3} inner target within 1000 iterations (statistics in `metadata.json`); the converged-step fraction is reported honestly in table 2.